

Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; CV, column volume; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSM, response-surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Protein A chromatography is the capture step of the A-Mab drug-substance process and the first chromatographic unit operation in the purification train. The step binds the antibody from clarified harvest on an immobilised Protein A resin, washes the bound product, and elutes it at low pH into the viral inactivation step that follows. It sets one quality attribute of its own, leached Protein A, and it is the principal clearance step for host cell protein (HCP) and residual DNA. The studies reported here executed the design prospectively defined in the paired plan (PCP-005) on a qualified scale-down model, under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012), and follow the A-Mab case study for the process model and the criticality framework (CMC Biotech Working Group 2009).

The step carries 6 process parameters. Four of them were studied multivariately (protein load, elution buffer pH, load flow rate and end of pool collect) in a 19-run two-level screening design followed by a 28-run face-centred response-surface design. The remaining two, operating temperature and bed height, were assessed outside the multivariate design. 2 parameters are classified as well-controlled critical process parameters, 2 as key process parameters and 2 as general process parameters. No parameter at this step required designation as a critical process parameter.

Pool host cell protein and step yield are well described by the response-surface models, with coefficients of determination of 0.989 and 0.913. Pool host cell protein rises with protein load and falls as elution buffer pH rises, and the two factors interact. Leached Protein A behaves differently. No process parameter had a significant effect on it, and the fitted surface predicts a run it has not seen no better than the mean of the response does, so the study reports the attribute as robust to the operating parameters over the ranges studied and does not treat its model as predictive. All 47 pools assayed across the two designs met the acceptance criterion of 5 ppm, the highest single determination being 4.61 ppm.

At drug substance the three impurity attributes this step governs all clear their limits with margin. Host cell protein is the tightest of them, at a process capability index of 6.14 against a mean of 15.2 ng/mg and a limit of 100 ng/mg. The tightest capability in the drug-substance process is not at this step; it is 1.51, and it belongs to a viral clearance attribute characterized in PCR-008 and PCR-009. No viral clearance is claimed for Protein A chromatography, and no aggregate reduction is claimed either. The pool host cell protein handed forward is not judged against the drug-substance specification, because the polishing steps reported in PCR-007 and PCR-008 carry that reduction.

2 deviations were recorded during execution. Both were investigated, impact-assessed and resolved without changing any parameter classification or the operating region defined in §6 (both are described in §13). This report supports the transfer scope in PTP-001 and rolls up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized immunoglobulin G subclass 1 monoclonal antibody whose principal mechanism of action is antibody-dependent cellular cytotoxicity. Its drug substance is produced by fed-batch cell culture followed by a platform purification train of affinity capture, low-pH viral inactivation, two polishing chromatography steps, small-virus retentive filtration and formulation by ultrafiltration and diafiltration. The train and the principal role of each step are given below (Table 1.1).

Table 1.1: The A-Mab drug-substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Protein A chromatography is operated in bind and elute mode. Clarified harvest is loaded on an immobilised Protein A resin, which binds the antibody through its Fc region while the bulk of the process impurities (host cell protein, DNA and media-derived small molecules) pass to the flow-through and the wash. A low-pH elution buffer releases the product into a pool that feeds the low-pH viral inactivation step directly, so the elution condition is shared between two unit operations. Small quantities of Protein A ligand are released from the resin during elution and enter the pool, so the step sets the leached Protein A attribute and does not clear it.

The affinity mechanism is selective for the Fc region and is not sensitive to the glycan or charge state of the molecule. Glycosylation and charge variants are formed in the production bioreactor (PCR-003) and pass through capture unchanged, so they were not carried as responses in this study. Aggregate also passes through. Aggregate reduction is credited to cation exchange chromatography (PCR-007).

1.2 Objectives

The study had five objectives.

- Quantify the effect of each multivariate parameter on pool host cell protein, step yield and leached Protein A across the characterization ranges.
- Define the multivariate operating region within which the quality attribute the step sets remains within its acceptance criterion.
- Establish proven acceptable ranges for each governed attribute and parameter, both at set-point and with the other parameters varying within their normal operating ranges.
- Classify every parameter of the step against the decision logic for parameter classification (SOP-4001).
- Justify the Stage 2 ranges and the step's contribution to the drug-substance control strategy.

1.3 Regulatory and scientific basis

The report is a Stage 1 process design record under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 for design space and parameter criticality, ICH Q9 for the risk basis of the study scope, and ICH Q11 for the link between process understanding and the control strategy (International Council for Harmonisation 2009, 2023, 2012). Criticality is treated as a continuum and not as a binary classification, so an attribute of low to moderate criticality such as leached Protein A is characterized and controlled even though it is not a release-limiting attribute. The process model, the impurity behaviour of each unit operation and the criticality framework are taken from the A-Mab case study (CMC Biotech Working Group 2009).

ICH Q5A is not invoked for this step. Affinity capture can be shown to remove enveloped virus, but no viral clearance claim is made here, and the cumulative clearance claim rests on the low-pH hold, anion exchange and small-virus retentive filtration. Those three steps are reported in PCR-006, PCR-008 and PCR-009 and are consolidated in PCMR-001. The documents that frame or receive this report are listed below (Table 1.2).

Table 1.2: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The A-Mab purification process is a platform process with an extensive performance history. Affinity capture on Protein A resin has been used at commercial scale for three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and through A-Mab clinical supply, with essentially unchanged column chemistry, buffer system and cycle structure. That experience establishes the mechanism of the step and the qualitative direction of every parameter effect, so the purpose of this characterization was to confirm and bound those effects for A-Mab, not to discover them. Prior knowledge was used to set the factor ranges and the mechanistic expectations stated below; it was not used in place of data for any conclusion in §5.

Four mechanistic expectations framed the design. Higher protein load increases the mass of antibody bound per unit resin volume, and the host cell protein that co-adsorbs with the product and elutes with it (both to the ligand and to the antibody itself) is expected to rise in proportion, so pool host cell protein was expected to increase with load. A lower elution pH is a harsher desorption condition that releases more weakly associated host cell protein together with the product, so pool host cell protein was expected to fall as elution buffer pH rises. Higher load flow rate shortens the residence time available for binding and was expected to reduce step yield through early breakthrough, most severely when the column is also heavily loaded. Collecting further into the trailing edge of the elution peak recovers more product and was expected to raise step yield.

Leached Protein A was expected to be governed by the resin and its cycle history and not by the operating parameters. Ligand release from a Protein A resin is driven by hydrolysis and proteolysis of the immobilised ligand, which depend on the base matrix, the packing quality, the sanitisation regime and the number of cycles the column has seen. Those variables are controlled under the resin life-cycle procedure (SOP-2008) and were held constant across the study. Prior campaign experience with an alternate resin is recorded in RA-004, which concluded that independent characterization would be required for any resin change and that no bridging from platform data would be accepted.

2.2 Quality attributes in scope

The step sets one quality attribute of its own, and that attribute is listed below with its acceptance criterion and criticality (Table [2.1](#)).

Table 2.1: Quality attribute set by the Protein A step, with acceptance criterion and criticality assigned under the Tool #1 impact and uncertainty ranking.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0-5 ppm	L-M	16

Leached Protein A is of low to moderate criticality. It is a process-derived impurity with an immunogenicity rationale, and it has no direct efficacy or safety effect at the levels reached; its acceptance criterion of 5 ppm applies at drug substance. The attribute is introduced here and reduced by every subsequent chromatography step, so this report gives the level the capture pool delivers and defers the cumulative position to the capability section (§8) and to PCMR-001.

Two further attributes are governed but not set here. They are the process impurities the capture step removes (Table 2.2).

Table 2.2: Quality attributes formed upstream and cleared by the Protein A step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

Host cell protein is the larger of the two in characterization terms, because the capture step delivers the single largest reduction of the purification train and because the pool level it produces sets the impurity burden the polishing steps have to handle. Residual DNA is of very low criticality and clears to well below its limit across the train. Neither attribute is judged against its drug-substance criterion at the outlet of this step.

2.3 Risk-based prioritization and parameter selection

The scope of this study was set by the pre-characterization risk assessment RA-001. That assessment ranked every parameter of the step on its potential main effect on quality attributes and on its potential to interact with other parameters, and assigned a study type from the combined score. Parameters with the highest combined scores were assigned to the multivariate design, parameters with intermediate scores to univariate assessment, and parameters whose ranges are fixed by equipment design or by an existing platform claim to no new study.

Four parameters were assigned to the multivariate design. Protein load and elution buffer pH were ranked highest because prior knowledge links both to pool host cell protein and because they were expected to interact. Load flow rate and end of

pool collect were included because both act on step yield and because the flow rate effect was expected to depend on the load, which is an interaction a univariate study cannot resolve. Operating temperature and bed height were assigned to univariate assessment. Temperature acts on binding kinetics and is held within a narrow band by the chromatography chamber, and bed height is fixed at packing under SOP-2008 and verified before use (plate count and peak asymmetry).

The characterization ranges were set wider than the routine control ranges, at roughly twice to three times the normal operating range for each factor, so that the study would define an operating region with margin around the set-point instead of merely confirming the set-point itself. The wider ranges also support later movement within the design space without new studies. The ranges are given in the study design section (Table [4.1](#)).

3 Materials and methods

3.1 Scale-down model and its qualification

A laboratory-scale chromatography system was qualified as a model of the commercial-scale Protein A step under SOP-1001, and all data in this report were generated on it. The model was designed on the established principles of chromatography scaling. It uses the same resin lot family, the same bed height as the commercial column, the same linear flow velocities, and the same load, wash and elution volumes expressed in column volumes, so that residence time and the number of column volumes at every phase of the cycle are preserved across scales. Column efficiency was verified by plate count and peak asymmetry before each sequence of runs.

Qualification compared the scale-down model against commercial-equivalent operation at the set-point in triplicate, on the performance parameters that carry the step's claims (step yield, elution pool volume, pool host cell protein, residual DNA and leached Protein A). No statistically significant difference was found between scales on any of these, so the model is accepted as representative of the full-scale system for the purposes of this study. The qualification data are held in the scale-down model qualification record filed under SOP-1001 and are not reproduced here.

The warrant this places on the study is specific and worth stating in its bounds. Because residence time, column volumes and bed height are matched, effects that act through binding kinetics, mass transfer and peak position transfer between scales. Effects that would act through wall support, packing heterogeneity or column-diameter-dependent flow distribution are not reproduced at laboratory scale, and no claim in this report depends on them. The operating region defined in §6 is therefore a claim about the scale-down model, verified at commercial scale only at the set-point.

3.2 Operation

Each run followed the cycle defined in SOP-2007. The column was equilibrated, loaded with clarified harvest at the run's protein load and load flow rate, washed with the platform wash sequence, and eluted with the run's elution buffer at the assigned pH. Pool collection started on the ascending edge at the platform absorbance threshold and stopped at the run's end of pool collect, expressed in column volumes from the start of elution. The column was then stripped, sanitised and re-equilibrated under SOP-2008.

Inputs held to platform specification (resin lot, buffer compositions other than the elution pH under study, wash sequence, equilibration conditions and column packing

quality) were controlled as identity and quality attributes and were not characterized. A single clarified harvest pool at nominal titre and impurity burden was used as feed for all runs, so that the measured effects are attributable to the operating parameters alone. Feed-state variation was not a factor of this design, and §6 states the bound that follows from that choice.

3.3 Analytical methods

Every response was measured by a validated method. The controlled documents governing the step and its analytics are listed below (Table 3.1).

Table 3.1: Standard operating procedures and validated analytical methods applied to the Protein A characterization studies.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Pool host cell protein was determined by the platform host cell protein enzyme-linked immunosorbent assay (AMV-3012) and is reported per unit of product mass, in ng/mg. Leached Protein A was determined by a ligand-specific enzyme-linked immunosorbent assay (AMV-3016) and is reported in ppm relative to product mass. Step yield was calculated from product mass in the pool against product mass in the load, using protein concentration by absorbance at 280 nm (AMV-3019). Residual DNA was determined by quantitative polymerase chain reaction (AMV-3014) and size variants by size-exclusion chromatography (AMV-3011); neither was carried as a response of the multivariate design, for the reasons given in §5.3 and §2.2. Method performance for the assays that carry this step’s claims is summarised in Appendix D.

3.4 Sampling plan

Samples were drawn at the same points in every run. The load was sampled once for titre and impurity burden, the flow-through and the wash were sampled for mass balance, and the elution pool was sampled after mixing for all three responses. Strip fractions were retained but were not assayed as part of this study. The retained pool aliquots were held frozen so that any determination flagged during analytical review could be repeated on the same material (see DEV-005-02).

The screening design carried 3 centre-point replicates and the response-surface design carried 4. Centre-point runs were executed at the set-point of every factor, distributed through the run order and not grouped, so that the replicate variation reflects the full range of day-to-day operation and not a single session.

3.5 Statistical methods

All models were fitted on coded factors, with the set-point of each factor at coded zero and the edges of its characterization range at coded minus one and plus one. For this step the set-point of every multivariate factor coincides with the midpoint of its characterization range, so coded zero and the set-point are the same condition. The screening model is an ordinary least-squares fit of the main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in response across the full range of that factor. The response-surface model is the full quadratic in the same four factors.

Significance was assessed at the 5 % level. Model adequacy was judged on four statements taken together (the coefficient of determination, the adjusted coefficient of determination, the predicted coefficient of determination computed from the prediction error sum of squares, and the lack-of-fit test against the centre-point pure error). The predicted coefficient of determination is the decisive one for a predictive claim, because it measures how well the fitted surface predicts an observation that was not used to fit it. A response for which no factor is active and for which the lack-of-fit test is not significant is reported as robust, and its model is retained as evidence of that robustness rather than used for prediction.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train varying within its normal operating range. Capability is expressed as a one-sided process capability index against the relevant drug-substance limit. Proven acceptable ranges were computed from the fitted response-surface model in two forms, described in §7.

4 Study design

4.1 Factors, ranges and the knowledge space

The design covers the four parameters that the risk assessment linked to a quality attribute or to process performance, across ranges that extend beyond the normal operating range in both directions. The set-point, normal operating range and characterization range of every parameter of the step are given in Table 4.1, together with the classification each carries at the end of the study and the study type assigned to it by RA-001.

Table 4.1: Process parameters of the Protein A step, with set-point, normal operating range, characterization range, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	30	20-40	10-50	WC- CPP	multivari- ate
Elution buffer pH	pH	3.55	3.4-3.7	3.2-3.9	WC- CPP	multivari- ate
Load flow rate	cm/hr	200	150-250	100-300	KPP	multivari- ate
End of pool collect	CV	2.6	2.3-2.9	2-3.2	KPP	multivari- ate
Operating temperature	°C	22	18-26	15-30	GPP	univariate
Bed height	cm	20	15-25	10-30	GPP	univariate

The characterization range is the knowledge space edge for each parameter and is not itself a proven acceptable range; the ranges that were derived from the data are reported later (§7). Coded factor letters are used throughout the effect and coefficient tables, and Table 4.2 maps them to the parameters.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM

Code	Factor	Unit	Studied in
D	End of pool collect	CV	screening + RSM

Each range was set from a distinct argument. Protein load was studied from well below to well above the set-point, spanning conditions from a lightly loaded column to a load approaching the dynamic binding capacity of the resin, because the upper edge is where breakthrough and elevated pool host cell protein are both expected. Elution buffer pH was bounded below by the pH at which antibody precipitation becomes a risk in the pool and above by the pH at which product recovery begins to fall, so the studied range brackets the whole usable elution window. Load flow rate was studied over a range that halves and doubles the residence time at the set-point. End of pool collect was studied from a conservative early stop to a stop well into the trailing edge, which is the region where additional product recovery competes with additional impurity carry-over.

4.2 Screening design

The screening study was a two-level full factorial in the four multivariate factors, augmented with centre points, giving 19 runs in randomised order. A full factorial in four factors estimates every main effect and every two-factor interaction free of aliasing (resolution is not a constraint at this factor count), which is why no fractional design was used. The purpose of the screening study was to identify which factors are active on each response and in what direction, and the design carries no quadratic terms, so it cannot detect curvature. The screening model is therefore an identification model and not a predictive one. The design matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same four factors, with axial points at the faces of the cube and centre-point replicates, giving 28 runs. All four screening factors were carried forward. Two of them were active on pool host cell protein (protein load and elution buffer pH) and two on step yield, and retaining all four keeps the predictive model defined over the same region as the knowledge space in Table 4.1. The axial points at the face of the cube allow the quadratic terms to be estimated without extending the design beyond the characterization ranges. The response-surface model is the predictive model of this report and is the basis of the operating region in §6 and the proven acceptable ranges in §7. The design matrix and the measured responses are given in Appendix B.

4.4 Univariate assessment

Operating temperature and bed height were assessed one factor at a time over the ranges in Table 4.1 and were not carried into either multivariate design. Neither parameter is a factor of the response-surface model, so the operating region defined in §6 makes no claim about them, and their ranges rest on the univariate assessment together with the operating and packing controls for the step (SOP-2007 and SOP-2008). Both remain general process parameters.

5 Results

5.1 Centre-point performance and reproducibility

The centre-point replicates of the response-surface design establish the pure-error term against which every lack-of-fit test in this section is judged, and they separate the three responses sharply. Table 5.1 gives the mean, standard deviation and coefficient of variation of each response over the replicates.

Table 5.1: Centre-point performance of the response-surface design, over replicate runs executed at the set-point of every factor.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7564	356.7	4.717
Step yield	4	0.9463	0.01865	1.971
Leached Protein A (ppm)	4	2.857	0.9669	33.84

Step yield is the most reproducible response, at a coefficient of variation of 2.0 %, and pool host cell protein follows at 4.7 %. Leached Protein A is the least precise by a wide margin, at 33.8 % about a mean of 2.86 ppm. That figure is a property of the attribute and its assay together, and §5.3 separates the two contributions. The screening design was less reproducible on pool host cell protein than the response-surface design (19.9 % against 4.7 %), which is consistent with the smaller number of replicates it carried.

5.2 Screening: factor effects

The screening study identified two active factors on pool host cell protein and three on step yield, and none on leached Protein A. Table 5.2 summarises the fit of the screening model to each response.

Table 5.2: Screening-model fit summary for the three measured responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.9728	0.9389	0.8861	28.65	3.492e-05	1400
Step yield	19	0.9756	0.9452	0.7977	32.04	2.279e-05	0.008329

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Leached Protein A (ppm)	19	0.3714	-0.4143	-2.981	0.4727	0.8674	0.5126

Pool host cell protein is dominated by elution buffer pH and protein load (coded factors B and A). Raising elution buffer pH across its range lowers pool host cell protein by 8,840 ng/mg, and raising protein load across its range increases it by 7,269 ng/mg. A significant interaction between the two was identified. No other term reaches significance, and load flow rate is the smallest effect of the four. The largest effects are given in Table 5.3.

Table 5.3: Largest screening effects on pool host cell protein. Effects are expressed as the change in response across the full characterization range of the factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8840	-4420	350	-12.63	1.451e-06	***
A	7269	3635	350	10.38	6.398e-06	***
AB	-2587	-1293	350	-3.695	0.006083	**
AD	1133	566.4	350	1.618	0.1442	
AC	830.1	415.1	350	1.186	0.2697	

Step yield is dominated by end of pool collect, which raises yield by 6.0% across its range. Protein load and load flow rate both reduce yield, and their interaction is significant, which is the expected signature of a capacity limit that only bites when a heavily loaded column is also run fast (§5.4). The three main effects and the interaction are given in Table 5.4.

Table 5.4: Largest screening effects on step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.06038	0.03019	0.002082	14.5	5.01e-07	***
A	-0.02553	-0.01277	0.002082	-6.131	0.0002797	***
C	-0.02367	-0.01183	0.002082	-5.684	0.0004629	***
AC	-0.02086	-0.01043	0.002082	-5.009	0.001041	**
CD	-0.01075	-0.005374	0.002082	-2.581	0.03256	*

Leached Protein A showed no significant effect from any factor or interaction. The screening model is not significant as a whole ($p = 0.87$), its adjusted coefficient of determination is negative, and the smallest p-value among the ten terms is 0.24. The full effect table for this response is given in Appendix C (Table 13.6). That null result was carried forward and not treated as an absence of data, and §5.3 reports what the response-surface design added to it.

Effect magnitudes from the screening study are estimates over a design with no curvature terms. Where the response-surface model resolves curvature, as it does for

elution buffer pH on pool host cell protein, the screening estimate of a main effect is an average over the range and understates the local slope at one end of it. The values quoted in the remainder of this report are therefore the response-surface estimates.

5.3 Response-surface models

Two of the three responses are adequately modelled and one is not, and the distinction governs how each is used in the rest of this report. Table 5.5 gives the fit of the response-surface model to each response.

Table 5.5: Response-surface model fit summary for the three measured responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.9887	0.9764	0.9319	80.94	2.713e-10	895
Step yield	28	0.9126	0.8184	0.5861	9.693	0.0001046	0.01602
Leached Protein A (ppm)	28	0.5671	0.1008	-0.7139	1.216	0.3649	0.6368

Pool host cell protein is well described. The model explains 0.989 of the variation with an adjusted value of 0.976, and its predicted coefficient of determination is 0.932, which is close enough to the adjusted value to support prediction rather than description alone. The lack-of-fit test is not significant at the 5 % level ($p = 0.058$), although the margin is narrow, so the quadratic form is accepted as adequate while the model is not extrapolated beyond the characterization ranges. The coefficient table is given in Table 5.6 and the analysis of variance in Table 5.7.

Table 5.6: Response-surface coefficients for pool host cell protein, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7216	309.5	23.31	5.443e-12	***
A	3872	211	18.35	1.12e-10	***
B	-5245	211	-24.86	2.404e-12	***
C	-435.4	211	-2.064	0.05958	
D	-1055	211	-5.003	0.0002416	***
AB	-1900	223.8	-8.49	1.159e-06	***
AC	-429.1	223.8	-1.918	0.07739	
AD	-518.1	223.8	-2.315	0.03758	*
BC	348.2	223.8	1.556	0.1437	
BD	727.4	223.8	3.251	0.006317	**
CD	-128.7	223.8	-0.5753	0.5749	
A ²	-344.5	557.3	-0.6181	0.5472	
B ²	1944	557.3	3.488	0.004009	**
C ²	851.3	557.3	1.528	0.1506	

Term	Coef.	Std. err.	t	p-value	Sig.
D ²	330.5	557.3	0.593	0.5634	

Table 5.7: Analysis of variance for the pool host cell protein model, with the residual partitioned into lack of fit and centre-point pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05	nan	nan

Four terms carry the pool host cell protein surface. Elution buffer pH and protein load are the two large main effects and act in opposite directions. Their interaction is significant, so the benefit of raising elution pH is larger at high load than at low load. The quadratic term in elution buffer pH is significant and positive, so the response flattens as elution pH approaches the top of its range (Figure 5.3). End of pool collect carries a small significant negative main effect that the screening design did not resolve. Pool host cell protein is expressed per unit of product mass, and collecting further into the trailing edge adds product mass to the denominator, which is consistent with the yield result for the same factor.

Step yield is adequately modelled, at a coefficient of determination of 0.913 and an adjusted value of 0.818, with no significant lack of fit ($p = 0.73$). Its predicted coefficient of determination is 0.586, which is materially lower than the adjusted value, so the model is used to describe the direction and rough size of yield changes and not to predict a single run. End of pool collect remains the dominant term, protein load and load flow rate remain negative, and the interaction between load and flow rate remains significant. The coefficient table is in Appendix C.

No process parameter had a significant effect on leached Protein A. The response-surface model is not significant as a whole ($p = 0.36$), its adjusted coefficient of determination is 0.101, and its predicted coefficient of determination is -0.714. A negative predicted coefficient of determination means the fitted surface predicts a held-out run less well than the mean of the response does. The lack-of-fit test is not significant ($p = 0.95$), so the residual variation is consistent with pure error and there is no evidence of structure the quadratic form has missed. One term (the quadratic in elution buffer pH) reaches nominal significance, and among the fourteen terms tested at the 5 % level that is what chance alone produces, so no weight is placed on it.

The variation that is present in this response is analytical and resin-derived. The centre-point replicates give a coefficient of variation of 33.8 % for leached Protein A, against 4.7 % for pool host cell protein over the same four runs. The leached Protein A assay (AMV-3016) has a precision of 6.5 % and accounts for 41 % of the observed variance in this response. The remainder is run-to-run variation in ligand release from

the resin itself, which is governed by the resin life cycle (packing, cycle number and sanitisation) and is controlled under SOP-2008.

This is a robustness result and it is reported as one. The response-surface model for leached Protein A is retained in the knowledge space as evidence that the attribute does not depend on protein load, elution buffer pH, load flow rate or end of pool collect over the ranges studied, and it is not used as a predictive model. No claim in §6 rests on it. The operative result for this attribute is the direct one: all 47 pools assayed across the screening and response-surface designs met the acceptance criterion of 5 ppm, and the highest single determination was 4.61 ppm.

Diagnostics for the two responses that carry the argument are given in Figure 5.1 and Figure 5.2. For pool host cell protein the residuals show no structure against the predicted value, the normal quantile plot is close to linear, and the actual against predicted plot lies on the identity line across the whole range of the response. For leached Protein A the residuals are equally free of structure, but the actual against predicted plot shows the predicted values compressed into a narrow band while the observations scatter across a much wider one, which is the visual form of a model with nothing to fit.

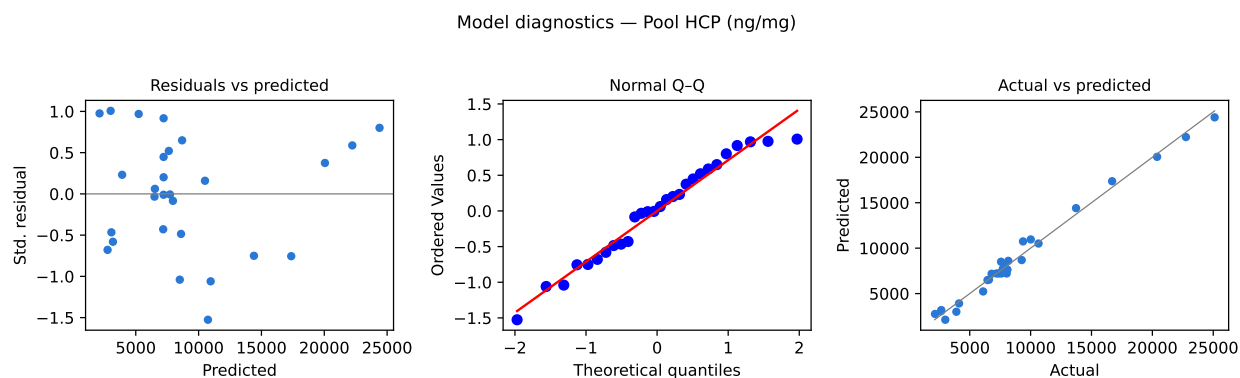


Figure 5.1: Model diagnostics for the pool host cell protein response-surface model: standardised residuals against predicted value, normal quantile plot, and actual against predicted.

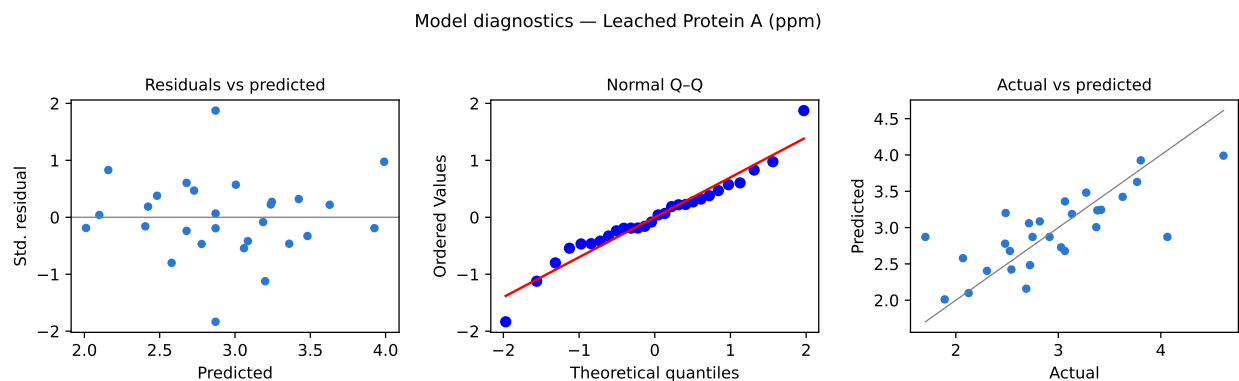


Figure 5.2: Model diagnostics for the leached Protein A response-surface model. The compressed spread of predicted values against the observed spread is the signature of a response with no active factor.

5.4 Mechanistic interpretation

The surfaces confirm the mechanistic expectations set out in §3.1, with one refinement. The predicted responses over protein load and elution buffer pH, with the other two factors at their set-points, are shown in Figure 5.3.

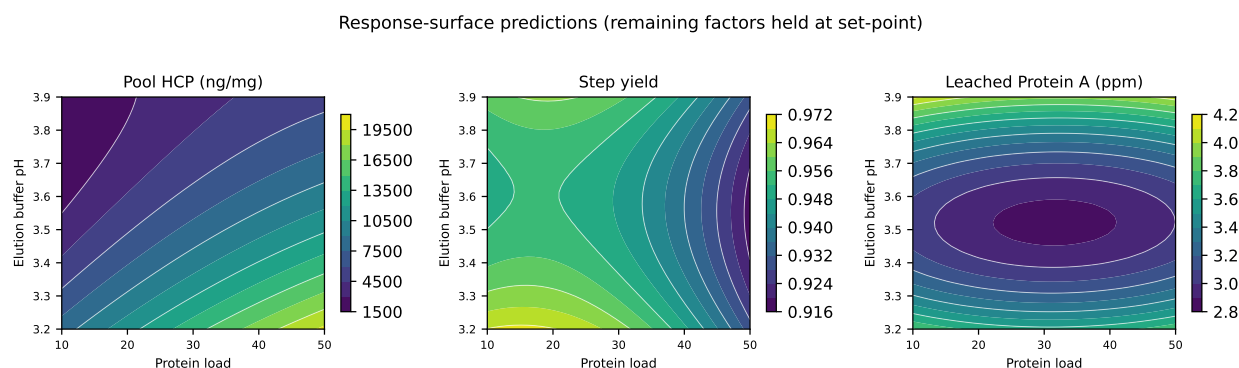


Figure 5.3: Response-surface predictions over protein load and elution buffer pH, with load flow rate and end of pool collect held at their set-points. Panels are pool host cell protein, step yield and leached Protein A.

Pool host cell protein varies over an order of magnitude across the characterization region. At the set-point the model predicts 7,216 ng/mg. At the corner that combines the highest protein load with the lowest elution buffer pH (the worst case for this response) it predicts 19,831 ng/mg, and at the opposite corner 1,598 ng/mg. Against a nominal clarified-harvest burden of 500,000 ng/mg, the step delivers 1.84 log reduction of host cell protein at the set-point and 1.40 log at that worst corner. Both directions follow from the adsorption mechanism. Host cell protein enters the pool by co-adsorption to the resin and to the product, so it rises with the mass of product bound,

and it is released preferentially at the harsher low-pH desorption condition. The interaction has the same origin: at low load there is spare ligand capacity and the pH of desorption matters less, while at high load the elution condition determines how much of the co-adsorbed impurity comes off with the product.

The curvature in elution buffer pH is the refinement prior knowledge did not supply. The positive quadratic term flattens the surface as elution pH rises, so the reduction in pool host cell protein per unit of pH is largest at the acidic end of the range and diminishes towards the top of it. Desorption of the weakly bound impurity is close to complete at the low end and the remaining impurity is more tightly associated, so further raising the elution pH buys progressively less. Operating above the set-point therefore gains less than a linear reading of the screening effect would suggest.

Step yield behaves as a capacity and recovery balance. Collecting further into the trailing edge recovers product that would otherwise be lost to the strip, which is the dominant term. Protein load and load flow rate each reduce yield by shortening the effective residence time per unit of product presented to the resin, and their significant interaction reflects that the two act on the same quantity. At the corner combining the highest load, the highest flow rate and the earliest pool stop, the model predicts a step yield of 84.4%, against 94.7% at the set-point.

These interactions are the reason the operating region is defined multivariately and not as a set of independent ranges. The acceptable elution pH depends on the protein load, and the acceptable load flow rate depends on the load, so a set of one-dimensional ranges would either be too wide at one corner or needlessly narrow everywhere else.

6 Design space

The design space for this step is the multivariate region in protein load, elution buffer pH, load flow rate and end of pool collect over which both impurity attributes the step governs remain within their in-process limits. That region is 73 % of the characterized region, evaluated on an evenly spaced grid over the four factors. Leached Protein A does not bound it. What bounds it is the pool host cell protein handed forward, which alone rejects 27 % of the grid.

The pool host cell protein level is not judged against the drug-substance specification at the outlet of this step, and no such claim is made here. Cation exchange (PCR-007) and anion exchange (PCR-008) reduce host cell protein to consistent levels well below the drug-substance limit, so the criterion that applies to the capture pool is an in-process limit of 10,724 ng/mg. It is the drug-substance limit carried back through the clearance those two steps deliver, divided by an assurance margin, and the cumulative position is consolidated in PCMR-001.

Over the normal operating ranges of the four parameters the model predicts pool host cell protein between 3,695 and 13,553 ng/mg, and step yield between 90.8% and 97.2%. The worst case the region must satisfy is the corner at maximum protein load and minimum elution buffer pH, where pool host cell protein is highest and the load presented to cation exchange is correspondingly highest. That corner is inside the characterized region and was executed in both designs (Table 13.2 and Table 13.3), and it sits outside the design space because the predicted pool concentration there is above the in-process limit.

The normal operating range of each parameter sits inside the characterized region, which in turn sits inside the knowledge space bounded by the characterization ranges in Table 4.1. Movement within the design space is not a change from a regulatory filing perspective under ICH Q8(R2) (International Council for Harmonisation 2009), although any planned movement is assessed prospectively under the quality system in line with ICH Q10 (International Council for Harmonisation 2008). Operation outside the characterized region requires change control and, where the mechanism is not already covered, new data.

Three bounds apply to this claim. The models cover the four multivariate parameters over the ranges in Table 4.1 and cover no other parameter (operating temperature and bed height in particular). All runs used a single clarified-harvest pool at nominal titre and impurity burden, so the region is defined for that feed state and the study does not bound the effect of feed variability. The region is defined on the qualified scale-down model and is not confirmed at commercial scale at its edges.

7 Proven acceptable ranges

The acceptance criteria applied in this section are the criteria that apply to this pool, and for the two impurity attributes the step governs they come from different places. For pool host cell protein the criterion is the in-process limit of 10,724 ng/mg, which is the drug-substance limit of 100 ng/mg carried back through the clearance that cation exchange and anion exchange deliver and then divided by an assurance margin. For leached Protein A the criterion is the drug-substance limit of 5 ppm itself. That is not an oversight. The backward calculation is not informative for leached Protein A, because the two polishing steps clear it by three orders of magnitude, so the drug-substance limit is already the tighter of the two at this pool and it is the one applied. Step yield is a process performance attribute with no acceptance criterion of this kind and is not carried in the proven acceptable range analysis. Two analyses were run for every combination of attribute and parameter, the first with the other three factors held at their set-points and the second with those factors varying within their normal operating ranges, propagated by Monte-Carlo simulation of the fitted response-surface model. The second analysis is the robustness form and is the reproducible default. Both are given below (Table 7.1).

Table 7.1: Proven acceptable ranges by governed attribute and parameter, at set-point and with the remaining factors propagated across their normal operating ranges.

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Protein load	10-50	g/L resin	$\leq$ 1.07e+04	10-49.8	10-34.5
Pool HCP (ng/mg)	Elution buffer pH	3.2-3.9	pH	$\leq$ 1.07e+04	3.36-3.9	3.48-3.9
Pool HCP (ng/mg)	Load flow rate	100-300	cm/hr	$\leq$ 1.07e+04	100-300	138-300
Pool HCP (ng/mg)	End of pool collect	2-3.2	CV	$\leq$ 1.07e+04	2-3.2	2.29-3.2
Leached Protein A (ppm)	Protein load	10-50	g/L resin	$\leq$ 5	10-50	10-50
Leached Protein A (ppm)	Elution buffer pH	3.2-3.9	pH	$\leq$ 5	3.2-3.9	3.2-3.86
Leached Protein A (ppm)	Load flow rate	100-300	cm/hr	$\leq$ 5	100-300	100-300

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Leached Protein A (ppm)	End of pool collect	2-3.2	CV	≤ 5	2-3.2	2-3.2

Host cell protein is a release attribute of the drug substance, and the drug substance does not exist until after anion exchange, so the drug-substance criterion is not what Table 7.1 measures the capture pool against. The in-process limit is, and against it every pool host cell protein row returns a range. Protein load and elution buffer pH are both bounded, at the top of the load range and at the bottom of the pH range, which are the two directions that raise the pool level. The measures that also apply to a capture step are the clearance factor it delivers and the consistency of the load it hands to the polishing steps. Both are given below for the nominal batch (Table 7.2).

Table 7.2: Host cell protein through the purification train for the nominal batch, with the clearance factor of each step and the document that reports it.

Stream	HCP (ng/mg)	Clearance factor (fold)	Reported in
Clarified harvest (Step 4)	500,000		PCR-004
Protein A pool (Step 5)	9,125	55	PCR-005
Cation exchange pool (Step 7)	117.0	78	PCR-007
Anion exchange pool (Step 8)	18.9	6.2	PCR-008

Capture reduces host cell protein by a factor of 55 from the clarified harvest, which is the largest single reduction in the train. Cation exchange (PCR-007) removes a further factor of 78 and anion exchange (PCR-008) a further 6.2, at which point the level is 18.9 ng/mg against the drug-substance limit of 100 ng/mg. The limit is met after anion exchange and not before. Since cation exchange and anion exchange reduce host cell protein to consistent levels well below that limit, the acceptable pool host cell protein from capture is linked to the operating conditions of those two steps, and the cumulative position is consolidated in PCMR-001.

What this study contributes is the quantified dependence of the capture pool level on the operating point, reported in §5.3 and §5.4. Over the normal operating ranges the fitted model predicts pool host cell protein between 3,695 and 13,553 ng/mg, a spread of about 3.7-fold, and the polishing steps are characterized against a load within that band. The same analysis is shown for the governing parameter in Figure 7.1, in the form used for the other attribute. No region is shaded in either panel, because the predicted response lies above the drug-substance limit across the whole range of elution buffer pH.

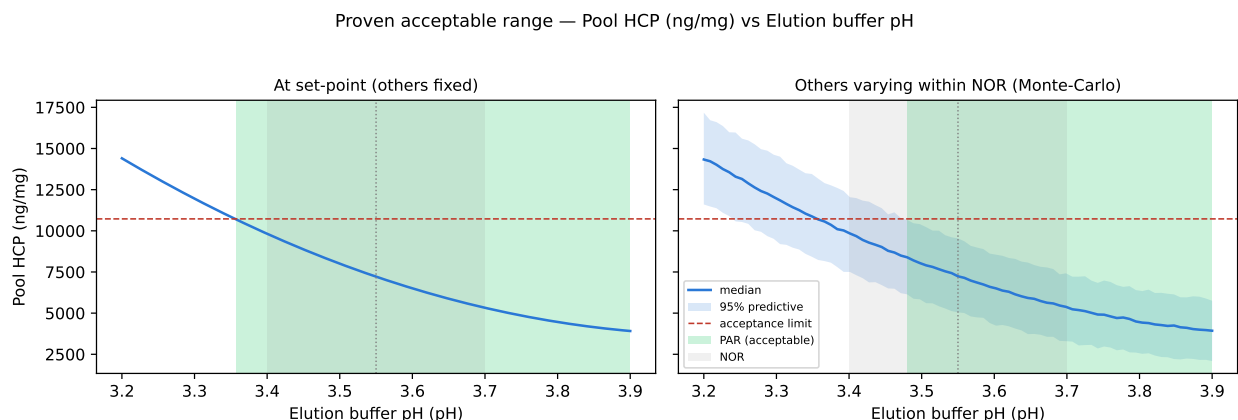


Figure 7.1: Proven acceptable range analysis for pool host cell protein against elution buffer pH, at set-point and with the remaining factors propagated across their normal operating ranges. The drawn limit is the in-process limit for this pool, and the acceptable region is shaded green.

Leached Protein A gives the opposite result. Its whole characterization range is proven acceptable for every parameter at set-point, and the propagation across the normal operating ranges narrows only one of them. For elution buffer pH the propagated range stops at 3.86 instead of the top of the characterization range, because the upper tail of the predictive distribution reaches the acceptance criterion there. That bound comes from the quadratic term discussed in §5.3, which is the only term in the model to reach nominal significance and which the study does not regard as established. It is retained because the analysis is run the same way for every response, and it is conservative. It also lies well above the upper edge of the normal operating range for elution buffer pH. The analysis for protein load (Figure 7.2), the governing parameter for this attribute, shows the acceptable region covering the whole characterization range in both panels.

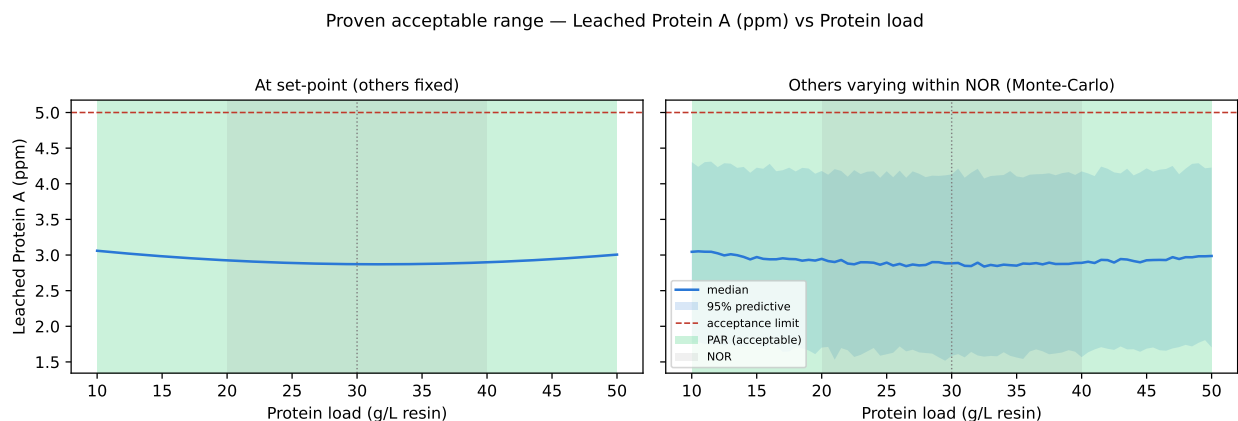


Figure 7.2: Proven acceptable range analysis for leached Protein A against protein load, at set-point and with the remaining factors propagated across their normal operating ranges. The acceptable region covers the whole characterization range in both panels.

The relationship between these ranges and the design space in §6 is a matter of dimension. The proven acceptable ranges are one-dimensional statements about a single parameter, taken either at set-point or with the other factors moving within their ranges, whereas the design space is the multivariate region in all four parameters at once. Where the whole characterization range of a parameter is proven acceptable, as it is for leached Protein A, the binding constraint on the operating region is not that parameter's univariate range. For this step the binding constraint is the multivariate pool host cell protein corner identified in §6, and it is set by the requirements of the downstream polishing steps.

Both analyses are bounded by the fitted model and by the characterization ranges. Neither extends to feed states other than the one studied, and neither covers operating temperature or bed height.

8 Process capability and robustness

Commercial-scale capability was estimated from a Monte-Carlo simulation of 2,000 batches with every parameter of the process train varying within its normal operating range. The result for the three attributes this step governs, at drug substance, is given below (Table 8.1).

Table 8.1: Commercial-scale capability at drug substance for the quality attributes governed by the Protein A step.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 $\pm$ 4.6	6.143
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.81
Leached Protein A	L-M	upper	0-5	0.003 $\pm$ 0.0006	2785

Host cell protein is the tightest of the three, at a one-sided capability index of 6.14, with a simulated mean of 15.2 ng/mg against a limit of 100 ng/mg. That margin is the combined result of three clearance steps and not of this one. Capture delivers 1.84 log of the reduction at the set-point, and the remainder comes from cation exchange (PCR-007) and anion exchange (PCR-008). Residual DNA clears with a wider margin, at a capability index of 10.81, and this step is credited with 2.5 log of DNA clearance, established by spiked clearance runs outside the multivariate design and verified at drug substance by quantitative polymerase chain reaction (AMV-3014).

Leached Protein A reaches drug substance at a mean of 0.0030 ppm against the acceptance criterion of 5 ppm. The capability index for this attribute is very large and carries no discriminating information, because the mean is orders of magnitude below the limit. The informative statement is the clearance: the capture pool carries 2.86 ppm at the set-point, and the downstream train removes about 3.0 log of it. Cation exchange and anion exchange carry that reduction and report it in PCR-007 and PCR-008.

None of these capabilities is the limiting one for the drug substance. The minimum capability index across all attributes of the process is 1.51, and it belongs to a viral clearance attribute characterized in PCR-008 and PCR-009. The capability figures in Table 8.1 are conditioned on operation within the normal operating ranges of every parameter of the train and on the scale-down models used to derive the underlying response models. They are not a statement about operation at the edges of the characterized region.

9 Parameter classification

Every parameter of the step was classified under the decision logic in SOP-4001, from the effects demonstrated in §5 and from the risk of the parameter leaving its range in routine operation. The classifications are carried in Table 4.1 and the reasoning for each is given below.

- Protein load is a well-controlled critical process parameter. It carries the second largest effect on pool host cell protein and a significant negative effect on step yield, so it is linked to a quality attribute of the drug substance. It is set by the mass loaded onto a column of known volume, which is directly measured and readily controlled, so the risk of operating outside the design space is low.
- Elution buffer pH is a well-controlled critical process parameter. It carries the largest single effect on pool host cell protein and interacts with protein load. It is controlled by buffer preparation and confirmed by release testing before use, which is a measurement with an established limit, so the risk of an undetected excursion is low (see DEV-005-01).
- Load flow rate is a key process parameter. It had no significant effect on either quality attribute in scope and a small negative effect on step yield, which places it in process performance and not in product quality.
- End of pool collect is a key process parameter. It carries the largest effect on step yield and a small significant effect on pool host cell protein through the normalisation of that response to product mass. It has no effect on the attribute this step sets, and the pool host cell protein it influences is not judged against a step-level criterion, so it does not meet the definition of a critical process parameter.
- Operating temperature is a general process parameter. It was not a factor of the multivariate design, it is held within a narrow band by the chromatography chamber, and RA-001 ranked it below the threshold for multivariate study.
- Bed height is a general process parameter. It is fixed at column packing under SOP-2008 and verified before use (plate count and peak asymmetry), so it is not an operating variable in routine manufacture.

No parameter of this step required designation as a critical process parameter. That conclusion rests on two findings taken together. The only quality attribute the step sets is insensitive to all four multivariate parameters over the ranges studied, and the attribute that does respond strongly to them is one whose drug-substance level is set by later steps. The drug-substance process carries 1 critical process parameter in total, at the low-pH viral inactivation step, and it is reported in PCR-006. The two factors with no significant effect on any quality attribute were classified and retained in the knowledge space, because their demonstrated inactivity is the evidence that supports the width of their ranges.

10 Contribution to the control strategy

The Protein A step contributes four elements to the drug-substance control strategy, and each follows from a result in §5.

Protein load and elution buffer pH are controlled within the normal operating ranges in Table 4.1, as the two parameters linked to a quality attribute. Elution buffer pH is additionally controlled at source: every elution buffer lot is released against a pH acceptance limit before use, and a lot outside that limit is not released. The sensitivity that justifies this control is quantified in the deviation record (§13). Load flow rate and end of pool collect are controlled within their ranges as key process parameters, to keep step yield and pool volume consistent, and they are monitored batch to batch as process attributes.

Pool host cell protein and leached Protein A are monitored on every batch as in-process attributes of the capture pool. Their function is trend detection and not acceptance, since neither is judged against the drug-substance specification at this step, and both are reported at drug substance under the release testing that supports PCMR-001. The resin is controlled through its life cycle under SOP-2008 (packing qualification, sanitisation and a maximum cycle number), which is the control that governs leached Protein A given the null result reported above (§5.3).

The step's contribution to impurity clearance is credited as follows. Capture delivers 1.84 log of host cell protein reduction and 2.5 log of DNA reduction at the set-point, and the drug-substance levels of both are achieved with cation exchange (PCR-007) and anion exchange (PCR-008). The step does not by itself assure any drug-substance impurity limit. It makes no viral clearance claim and no aggregate clearance claim, and it does not modify the glycan or charge-variant attributes formed in the bioreactor (PCR-003). The cumulative position across all steps is consolidated in PCMR-001.

11 Discussion

The step is well understood, and the understanding is asymmetric across the three responses in a way that is useful. Pool host cell protein has a predictive model over the four multivariate parameters, so the impurity burden handed to the polishing steps can be predicted from the operating point. Step yield has an adequate descriptive model that identifies the corner where recovery falls. Leached Protein A has no model, and that is the result rather than a gap in the study, because the attribute the step sets does not respond to the parameters the step controls.

The robustness of leached Protein A deserves its bounds stated plainly. The finding holds over the characterization ranges of the four multivariate parameters, on a single resin lot family operated within the cycle limits of SOP-2008, and with the assay (AMV-3016) contributing 41 % of the observed variance. It does not extend to a different resin, and RA-004 has already determined that a resin change would require independent characterization with no bridging from platform data. It also does not extend beyond the cycle limit, since ligand release is known to increase with cycle number and cycle number was held constant here.

Confidence in the scale-down model is high for the effects this study reports and is limited by design. Residence time, column volumes and bed height are matched between scales, so the adsorption and desorption effects that drive every result in §5 transfer. Effects that would arise from packing heterogeneity at scale, from wall support or from flow distribution across the column diameter are not reproduced, and no claim in this report depends on them. The models predict mean levels and carry no interval of their own, which is why the proven acceptable range analysis in §7 uses a propagated predictive distribution and not the fitted mean alone.

Two deviations were recorded and both were retained after impact assessment. The first was a buffer preparation error caught by release testing before the material reached a column, and the second was an analytical result that could not be confirmed either way, dispositioned conservatively by retaining the original determination. Neither changed a parameter classification, an effect estimate or the operating region. Both are described in §13.

The principal limitations of the study are three. All runs used one clarified-harvest feed state, so the study bounds the effect of the operating parameters and not of feed variability; the A-Mab platform history supports robustness to feed variation, but this study does not add to it. The operating region is defined on a laboratory-scale model and is not confirmed at commercial scale at its edges. Resin cycle number was held constant, so the life-cycle contribution to leached Protein A is controlled by SOP-2008 and validated separately, and it is not characterized here.

12 Conclusions

The Protein A capture step is well characterized and robust across the ranges studied. A response-surface model over the four multivariate parameters predicts pool host cell protein across the characterized region, and a second model describes step yield. Leached Protein A, the only quality attribute the step sets, showed no dependence on any of the four parameters, and all 47 pools assayed met the acceptance criterion of 5 ppm.

The operating region is defined multivariately, with the normal operating ranges of Table 4.1 nested inside it. 2 parameters are well-controlled critical process parameters, 2 are key process parameters and 2 are general process parameters, and no parameter required designation as a critical process parameter. At drug substance the three impurity attributes the step governs meet their limits, with host cell protein the tightest at a capability index of 6.14.

The step makes no viral clearance claim, no aggregate clearance claim, and no claim that it delivers any drug-substance impurity limit on its own. The 2 recorded deviations were resolved without changing any classification or the operating region. The step is ready for Stage 2 at the ranges in Table 4.1, and this report rolls up into PCMR-001.

13 Deviations from the plan

2 deviations from PCP-005 were recorded during execution. Each was investigated, impact-assessed and dispositioned before the data were analysed, and neither altered a parameter classification, an effect estimate or the operating region defined in §6. The register is given in Table 13.1.

Table 13.1: Register of deviations recorded during execution of the Protein A characterization studies.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	buffer release testing	retained
DEV-005-02	Leached Protein A result initially out-of-trend; re-assayed	analytical review	retained

13.1 DEV-005-01 — elution buffer prepared below target

An elution buffer lot (LOT-BUF-5290) was prepared for a response-surface run and released at pH 3.38, below its preparation target and below the lower edge of the normal operating range for elution buffer pH. The magnitude matters at this step more than it would elsewhere, because elution buffer pH is the largest single effect on pool host cell protein. Had the lot been used with the other factors at their set-points, the fitted model predicts a pool host cell protein of 10,221 ng/mg against 7,216 ng/mg at target, a factor of 1.42.

The deviation was detected at buffer release testing, before the lot was issued to the column. Investigation by the buffer preparation area identified an endpoint error during titration as the root cause, with no equipment fault and no other lot affected. The non-conforming lot was rejected, a replacement lot was prepared and released within its pH acceptance limit, and the affected run was executed with the replacement.

The impact on the study is therefore none. No non-conforming buffer reached a column, and the run in question was executed at the elution pH assigned to it in the design, which is the value shown in Appendix B (Table 13.3). The affected run and its data were retained. The corrective action was a revision to the buffer preparation record requiring a second independent pH verification before release (SOP-2103). The deviation also confirms the value of buffer pH release testing as a control-strategy element, and §10 records it as one.

13.2 DEV-005-02 — leached Protein A determination out of trend

One leached Protein A determination was flagged during analytical review as out of trend on the high side against the other runs of the same design. Analytical Development investigated the plate record and identified suspected carryover from an adjacent high-concentration well as the most likely cause, but the plate layout and the wash record could not establish it. The retained aliquot of the same pool was re-assayed by the same validated method (AMV-3016) and returned 1.6 ppm, below the centre-point mean of 2.86 ppm and well below the acceptance criterion of 5 ppm.

Because carryover was suspected but not confirmed, the original determination could not be invalidated, and it was retained in the reported data set. The disposition is deliberately conservative, because the higher of the two determinations is the one carried into every model and every table in §5. The re-assay was recorded in the analytical investigation and was not substituted for the original.

The deviation is consistent with the assay behaviour reported in §5.3. The leached Protein A method has a precision of 6.5 % and a limit of quantitation of 0.25 ppm, and it accounts for 41 % of the observed variance in this response. A single determination that separates from the others by more than the method's repeatability is expected at that variance, and it is one of the reasons the study does not read a factor effect into the leached Protein A surface. No classification, effect estimate or acceptance conclusion changed as a result. The corrective action was a revision to the plate layout in the analytical method record, placing a wash well between samples of widely different concentration.

Appendices A-D

Appendix A — Screening design matrix and responses

The coded screening design matrix and the measured responses are given in Table 13.2. Factor letters follow Table 4.2, and the coded levels are the edges and the set-point of each characterization range.

Table 13.2: Screening design matrix in coded factors, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	factorial	-1	-1	-1	-1	9,965.7	0.918	2.64
2	factorial	-1	-1	-1	1	9,515.4	0.99	3
3	factorial	-1	-1	1	-1	9,022.6	0.928	3.03
4	factorial	-1	-1	1	1	7,893.7	0.98	2.98
5	factorial	-1	1	-1	-1	3,211.7	0.915	2.88
6	factorial	-1	1	-1	1	2,523.1	0.986	3.82
7	factorial	-1	1	1	-1	2,540.2	0.929	3.3
8	factorial	-1	1	1	1	3,108.1	0.96	2.43
9	factorial	1	-1	-1	-1	16,568.2	0.913	2.47
10	factorial	1	-1	-1	1	19,737.9	0.99	2.66
11	factorial	1	-1	1	-1	18,271.3	0.857	2.67
12	factorial	1	-1	1	1	21,244.6	0.936	3.54
13	factorial	1	1	-1	-1	6,791.9	0.911	2.77
14	factorial	1	1	-1	1	7,875.0	0.976	2.29

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
15	facto- rial	1	1	1	-1	7,655.0	0.891	2.88
16	facto- rial	1	1	1	1	7,791.9	0.927	3.14
17	center	0	0	0	0	6,436.2	0.931	2.91
18	center	0	0	0	0	9,652.2	0.933	3.07
19	center	0	0	0	0	8,221.9	0.942	3.86

Appendix B — Response-surface design matrix and responses

The coded response-surface design matrix and the measured responses are given in Table 13.3. Axial runs sit at the faces of the cube, so no setting exceeds the characterization ranges.

Table 13.3: Response-surface design matrix in coded factors, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	facto- rial	-1	-1	-1	-1	10,011.3	0.93	3.42
2	facto- rial	-1	-1	-1	1	9,269.0	0.99	3.06
3	facto- rial	-1	-1	1	-1	10,650.9	0.928	2.53
4	facto- rial	-1	-1	1	1	7,715.2	0.988	2.48
5	facto- rial	-1	1	-1	-1	2,992.5	0.923	3.27
6	facto- rial	-1	1	-1	1	2,149.1	0.977	1.89
7	facto- rial	-1	1	1	-1	2,643.4	0.908	4.61
8	facto- rial	-1	1	1	1	2,663.4	0.99	3.13
9	facto- rial	1	-1	-1	-1	25,114.3	0.89	3.38
10	facto- rial	1	-1	-1	1	20,388.5	0.99	2.49
11	facto- rial	1	-1	1	-1	22,755.7	0.861	2.12
12	facto- rial	1	-1	1	1	16,694.1	0.924	3.03

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
13	facto- rial	1	1	-1	-1	7,882.2	0.895	3.63
14	facto- rial	1	1	-1	1	6,577.7	0.97	2.72
15	facto- rial	1	1	1	-1	6,798.5	0.844	3.06
16	facto- rial	1	1	1	1	6,100.0	0.962	2.82
17	axial	-1	0	0	0	3,900.5	0.935	2.71
18	axial	1	0	0	0	9,377.3	0.935	3.37
19	axial	0	-1	0	0	13,731.8	0.966	3.77
20	axial	0	1	0	0	4,121.9	0.955	3.8
21	axial	0	0	-1	0	7,571.5	0.955	2.3
22	axial	0	0	1	0	8,097.7	0.955	2.54
23	axial	0	0	0	-1	8,167.0	0.908	2.07
24	axial	0	0	0	1	6,460.6	0.953	2.69
25	center	0	0	0	0	8,036.2	0.94	2.75
26	center	0	0	0	0	7,616.5	0.967	1.7
27	center	0	0	0	0	7,206.4	0.924	4.06
28	center	0	0	0	0	7,396.1	0.954	2.91

Appendix C — Full effect and coefficient tables

The complete screening effect table for every response is given in Table 13.4 to Table 13.6, and the response-surface coefficient tables not shown in §5.3 are given in Table 13.7 and Table 13.8.

Table 13.4: Full screening effect table — pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8840	-4420	350	-12.63	1.451e-06	***
A	7269	3635	350	10.38	6.398e-06	***
AB	-2587	-1293	350	-3.695	0.006083	**
AD	1133	566.4	350	1.618	0.1442	
AC	830.1	415.1	350	1.186	0.2697	
D	707.9	353.9	350	1.011	0.3415	
BD	-433.1	-216.5	350	-0.6187	0.5533	
C	167.3	83.66	350	0.239	0.8171	
CD	-70.59	-35.3	350	-0.1008	0.9222	
BC	6.075	3.038	350	0.008679	0.9933	

Table 13.5: Full screening effect table — step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.06038	0.03019	0.002082	14.5	5.01e-07	***
A	-0.02553	-0.01277	0.002082	-6.131	0.0002797	***
C	-0.02367	-0.01183	0.002082	-5.684	0.0004629	***
AC	-0.02086	-0.01043	0.002082	-5.009	0.001041	**
CD	-0.01075	-0.005374	0.002082	-2.581	0.03256	*
BD	-0.009809	-0.004904	0.002082	-2.355	0.04629	*
AB	0.004426	0.002213	0.002082	1.063	0.3189	
AD	0.003702	0.001851	0.002082	0.889	0.3999	
BC	0.003499	0.001749	0.002082	0.8401	0.4252	
B	-0.002364	-0.001182	0.002082	-0.5677	0.5858	

Table 13.6: Full screening effect table — leached Protein A. No term is significant.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.3291	0.1646	0.1282	1.284	0.2351	
A	-0.208	-0.104	0.1282	-0.8115	0.4406	
BD	-0.1888	-0.09438	0.1282	-0.7365	0.4825	
BC	-0.1824	-0.09119	0.1282	-0.7116	0.497	
C	0.1796	0.08979	0.1282	0.7006	0.5034	
D	0.1516	0.07579	0.1282	0.5914	0.5706	
AB	-0.1329	-0.06643	0.1282	-0.5184	0.6182	
CD	-0.09871	-0.04936	0.1282	-0.3851	0.7102	
B	0.06523	0.03261	0.1282	0.2545	0.8055	
AD	0.05673	0.02837	0.1282	0.2214	0.8304	

Table 13.7: Response-surface coefficients — step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.9474	0.005538	171.1	3.503e-23	***
A	-0.01647	0.003775	-4.364	0.000767	***
B	-0.002427	0.003775	-0.643	0.5314	
C	-0.008909	0.003775	-2.36	0.03457	*
D	0.03654	0.003775	9.68	2.62e-07	***
AB	0.002664	0.004004	0.6654	0.5174	
AC	-0.00914	0.004004	-2.283	0.03991	*
AD	0.00629	0.004004	1.571	0.1402	
BC	0.002425	0.004004	0.6057	0.5552	
BD	0.002912	0.004004	0.7272	0.48	
CD	0.00219	0.004004	0.547	0.5937	
A ²	-0.01282	0.009972	-1.286	0.2209	
B ²	0.01195	0.009972	1.199	0.2521	

Term	Coef.	Std. err.	t	p-value	Sig.
C ²	0.006809	0.009972	0.6828	0.5067	
D ²	-0.0178	0.009972	-1.785	0.09764	

Table 13.8: Response-surface coefficients — leached Protein A.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.871	0.2202	13.04	7.67e-09	***
A	-0.02705	0.1501	-0.1802	0.8598	
B	0.1484	0.1501	0.9888	0.3408	
C	0.00918	0.1501	0.06116	0.9522	
D	-0.2102	0.1501	-1.401	0.1848	
AB	-0.01267	0.1592	-0.07958	0.9378	
AC	-0.1431	0.1592	-0.8992	0.3849	
AD	0.1323	0.1592	0.8313	0.4208	
BC	0.2689	0.1592	1.689	0.1151	
BD	-0.226	0.1592	-1.42	0.1793	
CD	0.1666	0.1592	1.046	0.3144	
A ²	0.1617	0.3965	0.4078	0.6901	
B ²	0.9062	0.3965	2.285	0.03972	*
C ²	-0.4576	0.3965	-1.154	0.2692	
D ²	-0.5022	0.3965	-1.267	0.2275	

Appendix D — Analytical methods summary

Performance data for the validated methods applied to the Protein A characterization studies are given in Table 13.9. Methods without performance figures in the table are covered by their own validation reports.

Table 13.9: Validated analytical methods applied to the Protein A characterization studies, with reported precision, limit of quantitation and accuracy.

Refer- ence	Method	Precision (%)	LOQ	LOQ unit	Accuracy (%)
AMV- 3016	Leached Protein A by ELISA (ppm)	6.5	0.25	ppm	97.4
AMV- 3012	Host-Cell Protein ELISA	9.5	1	ng/mg	95
AMV- 3014	Residual DNA (qPCR)				
AMV- 3011	Size-Variants (SEC-HPLC)				

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